

STAT 412 HW07 — Question 13

Problem

One research study of illegal drug use among 12- to 17-year-olds reported a decrease in use from 11.4% in 1997 to 9.7% now. Suppose a survey in a large high school reveals that, in a random sample of 1,050 students, 99 report using illegal drugs. Use a 0.05 significance level to test the principal's claim that illegal drug use in her school is below the current national average.

The conclusion choices include:

- A. Do not reject H_0 ; sufficient evidence to support the claim.
- B. Reject H_0 ; insufficient evidence to support the claim.
- C. Do not reject H_0 ; insufficient evidence to support the claim.
- D. Reject H_0 ; sufficient evidence to support the claim.

Solution

The principal's claim is "below the current national average," so this is a left-tailed test:

$$H_0 : p = 0.097, \quad H_a : p < 0.097.$$

This is choice D.

The sample proportion is

$$\hat{p} = \frac{99}{1050} \approx 0.0942857.$$

Using the exact sample proportion,

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.0942857 - 0.097}{\sqrt{0.097(0.903)/1050}} \approx -0.29718,$$

which rounds to -0.30 .

The platform-confirmed test statistic is $z = -0.30$.

The P-value for the found statistic is

$$P(Z < -0.30) = 0.3821.$$

Since $0.3821 > 0.05$, do not reject H_0 . The correct conclusion choice is C.

Final Answer

Choice D: $H_0 : p = 0.097$; $H_a : p < 0.097$. $z = -0.30$; P-value = 0.3821. Do not reject H_0 ; conclusion choice C.